



ENGINEERING THE FOUNDATION: THE RE-RASSOR ARM

*A SUMMARY OF THE PRELIMINARY DESIGN
PHASE & CURRENT PRODUCTION*

PREPARED BY TEAM L08
2026 RE-RASSOR SENIOR DESIGN TEAM

MEET THE TEAM

Max

CAD Dev

- Responsibilities:
- Designing & CADing the arm's base and connection mechanisms to the existing rover
 - Printing & Assembly of Components

Tolu

CAD Dev

- Responsibilities:
- Designing & CADing the arm's end effector and grabbing mechanism for the arm
 - Printing & Assembly of Components

Kaitlyn

Project Manager

- Responsibilities:
- Jira & Schedule Management
 - Assisting with CADing the center part of the arm & counterbalance system + printing responsibilities
 - Initial Stretch Goal Analysis

Lucas

Software Dev

- Responsibilities:
- Coding HTTP Requests for the arm
 - Integrating new requests with preexisting rover control systems

Nam

Software Dev

- Responsibilities:
- Coding the UI & API for the arm
 - Integrating new UI & API with preexisting rover control systems

Sponsored by The Florida Space Institute
Funded by the Steven Hawking Center for Microgravity Research

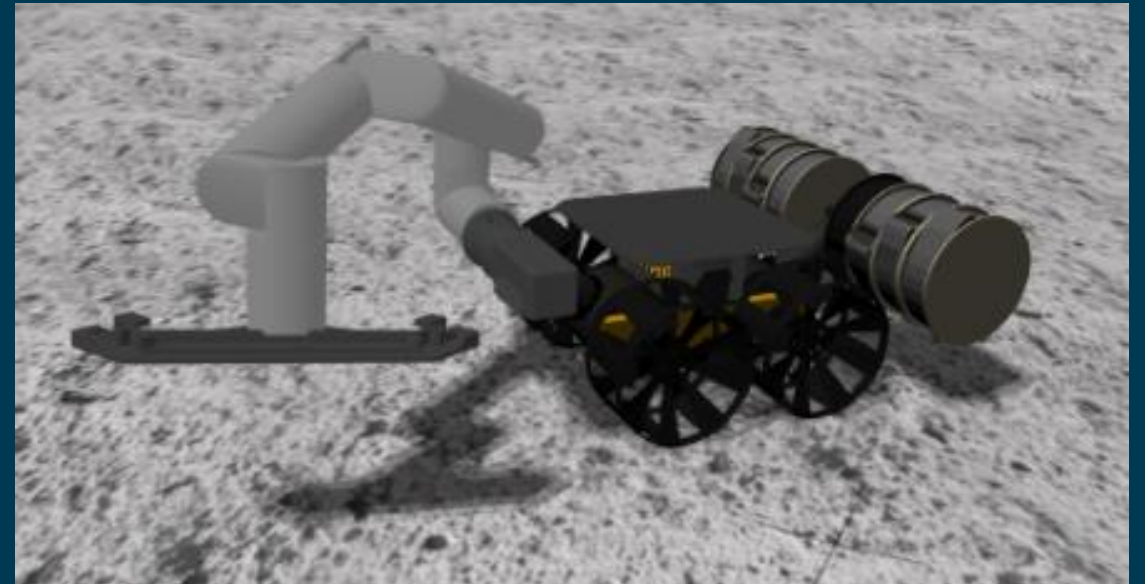
PROBLEM STATEMENTS

Problem 1:

The Florida Space Institute is exploring 3D printing space craft and that requires initial prototypes to prove feasibility. Previous arm attempts have proven unsuccessful due to structural failure from heavy components and failed to mount onto the rover chasses.

Problem 2:

Due to the longevity of this project the documentation is scattered among 12 repositories and there is no singular source for all current working components.



OUR SOLUTION

- Design a lightweight, modular robotic arm capable of manipulating standardized 3D-printed pavers based off of our rover arm's of similar scale and function utilizing the lighter brushless DC motor.
- Extend the existing software architecture through HTTP-based arm controls without disrupting the rover's current control system.
- Produce comprehensive engineering documentation to support long-term maintainability and knowledge transfer to future development teams.
- Organize as much of the existing documentation into our repository and the new Florida Space Institute Onshape account to allow subsequent teams to pick up where we left off

REQUIREMENTS

- **The Robotic Arm:** must be mechanically capable of targeting, lifting, moving, and precisely positioning the project's standard 3D printed pavers.
- **Fully Realized 3D Assembly Models:** STEP and .STL files detailing every link, joint, bracket, and gear assembly of the robotic arm.
- **Assembly Instructions:** Step by step documentation detailing physical assembly, fastener sizing, wire routing constraints, and maintenance requirements.
- **Extension of Existing Software Architecture:** Utilize the existing HTTP Request architecture to allow our arm to be controlled by similar logic to the drivetrain

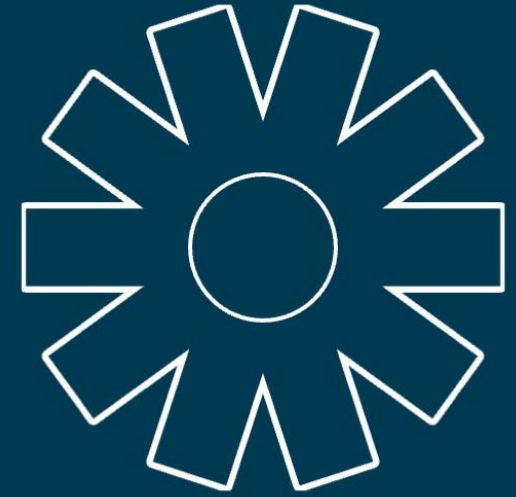
TECH STACK

Hardware:

- Printing Materials: PLA, PETG, Bambu X1-Carbon Printer
- Rover Components: BLDC/Stepper motors, Le Potato, Arduinos, BL Cooling Fan, Power Switch, 12.8V Battery, ArduCam
- Other Material Components: Netgear Router, Laptops

Software:

- Production: GitHub/Git, Arduino IDE, Android Studio, Visual Studio
- Libraries & Language: JavaScript, React, HTML, CSS, C++, Python
- CAD & Printing Software: Onshape, AutoCAD, Fusion 360, Bambu Studio



THE DEVELOPMENT PLAN

Physical Rover Plan

- Discover failure points of the arms to figure out the core design of our rover
- Front end designs and prints version 1 of arm in PLA. Then tests and fixes our major failure points.
- Version 2 (where the project is currently at) is developed and printed with PLA and final fixes are made after testing
- Final arm is printed in PETG & make final fixes

Software Plan

- Condense the previous 12 GitHub Repositories to get the scope of the current HTTP request and UI design
- Requests and UI are designed simultaneously to the physical rover and tested where possible on the prototypes

THE JIRA

Senior Design L08 RE-RASSOR

Summary

List

Board

Code

Forms

Timeline

Docs

+

Search board

KA

Filter

Group

Backlog & Blocked 4

Make Assemble Instructions & Video for V2 Prototype

Nov 9, 2026

☒ KAN-86

Instructor Demo

Oct 17, 2026

☒ KAN-87

Final Presentation (May be on the 30th depending on when we are assigned this is the saf...

Nov 16, 2026

☒ KAN-88

Update Design Doc for V2 Results

Oct 20, 2026

☒ KAN-94

+ Create

To Do 3

Validate & Test Complete Software Stack

Sep 30, 2026

☒ KAN-89

Validate V2 Prototype

Oct 16, 2026

☒ KAN-92

V2 Analysis & Testing Summary

Oct 19, 2026

☒ KAN-93

+ Create

In Research 2

Stretch Goal Evaluation

☒ KAN-112

CAD & Print V3 of End Effector

Oct 16, 2026

☒ KAN-118

+ Create

In Progress 2

Link UI to HTTP Requests

Sep 19, 2026

☒ KAN-82

Finalize arm control http requests

Sep 19, 2026

☒ KAN-51

+ Create

In Testing 3

Finalize arm control UI

Sep 19, 2026

☒ KAN-27

Bug Fixes For HTTP

Oct 23, 2026

☒ KAN-85

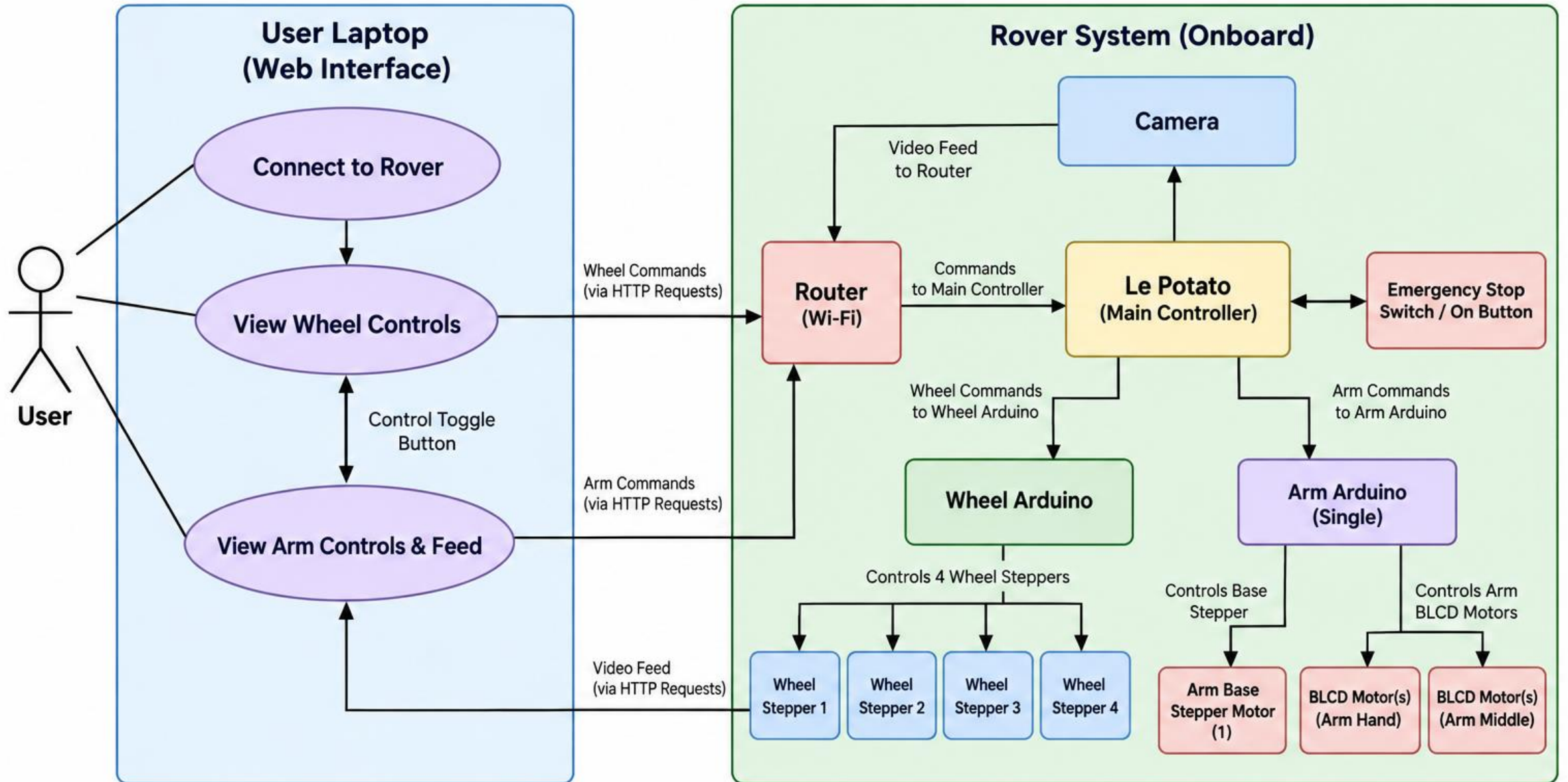
Bug Fixes for UI / API

Oct 23, 2026

☒ KAN-84

+ Create

USE CASE DIAGRAM



- 3x Brushless DC Motors - \$59
- 1x Nema 17 Stepper Motor - \$10
- 3 Rolls of PLA Filament for Testing - \$57
- Approx. 2 Rolls of PETG Filament - \$30
- 2x Arduino Mega 2560 - \$100
- 1x Libre Computer Le Potato - \$60
- 1x Nermak LiFePO₄ 12.8V Battery - \$120
- 1x NetGear Nighthawk Router. - \$70
- 1x 12V Brushless Cooling Fan. - \$15
- 1x CMORSUN BE102 Emergency Stop. - \$7
- Miscellaneous Mechanical Hardware (e.g. Bolts, nuts, etc.) - \$62

CURRENT MATERIALS & COST

**From Scratch Recreation Estimate:
\$590**

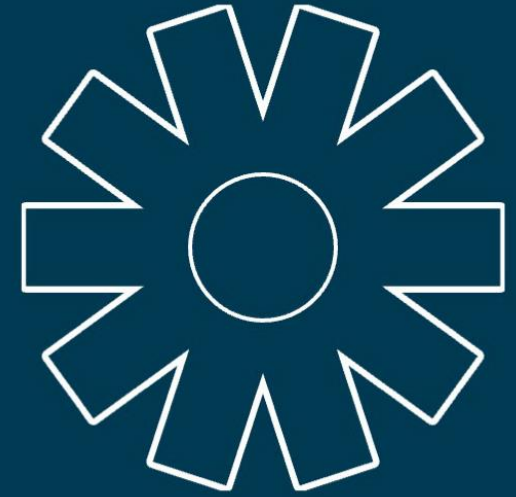
**Budget:
\$600-800**

**Spent By Team:
\$0**

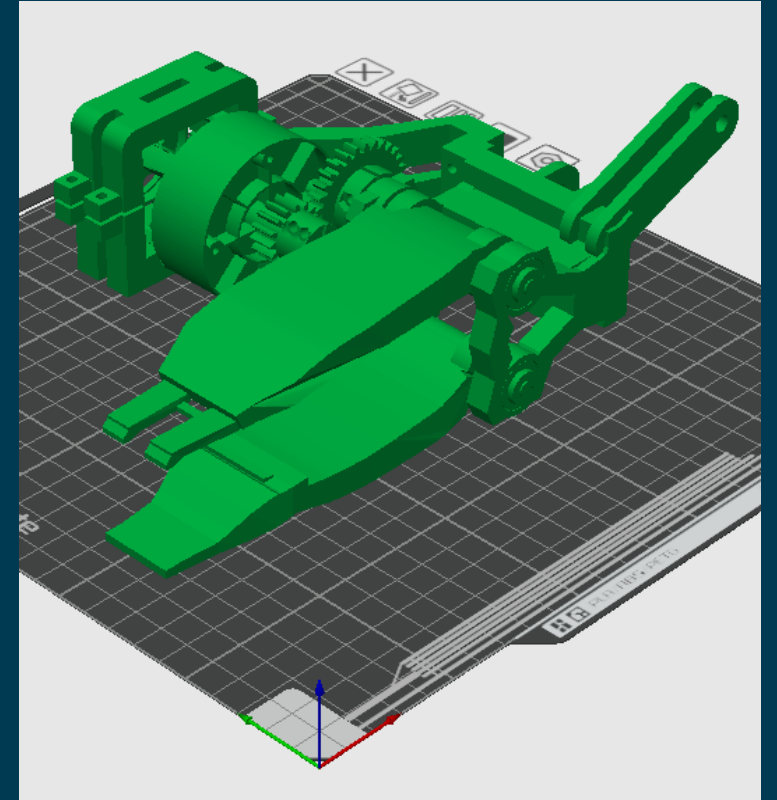
DESIGN		DECISION	MATRIX
Design Decision	Alternatives	Engineering Considerations	Final Selection
Actuation	Stepper / Servo / BLDC	Weight, torque, integration, efficiency	Majority BLDC With a Stepper Base
Arm Location	Front / Rear / Center Mount	Stability, workspace, compatibility	Front
Communication	HTTP / Serial / ROS	Existing infrastructure, simplicity	HTTP
Printing Material	PETG / PLA	Cost, safety, heat resistance	PETG for Final PLA to Test
			11

OUR STATUS REPORT & SUCCESSES

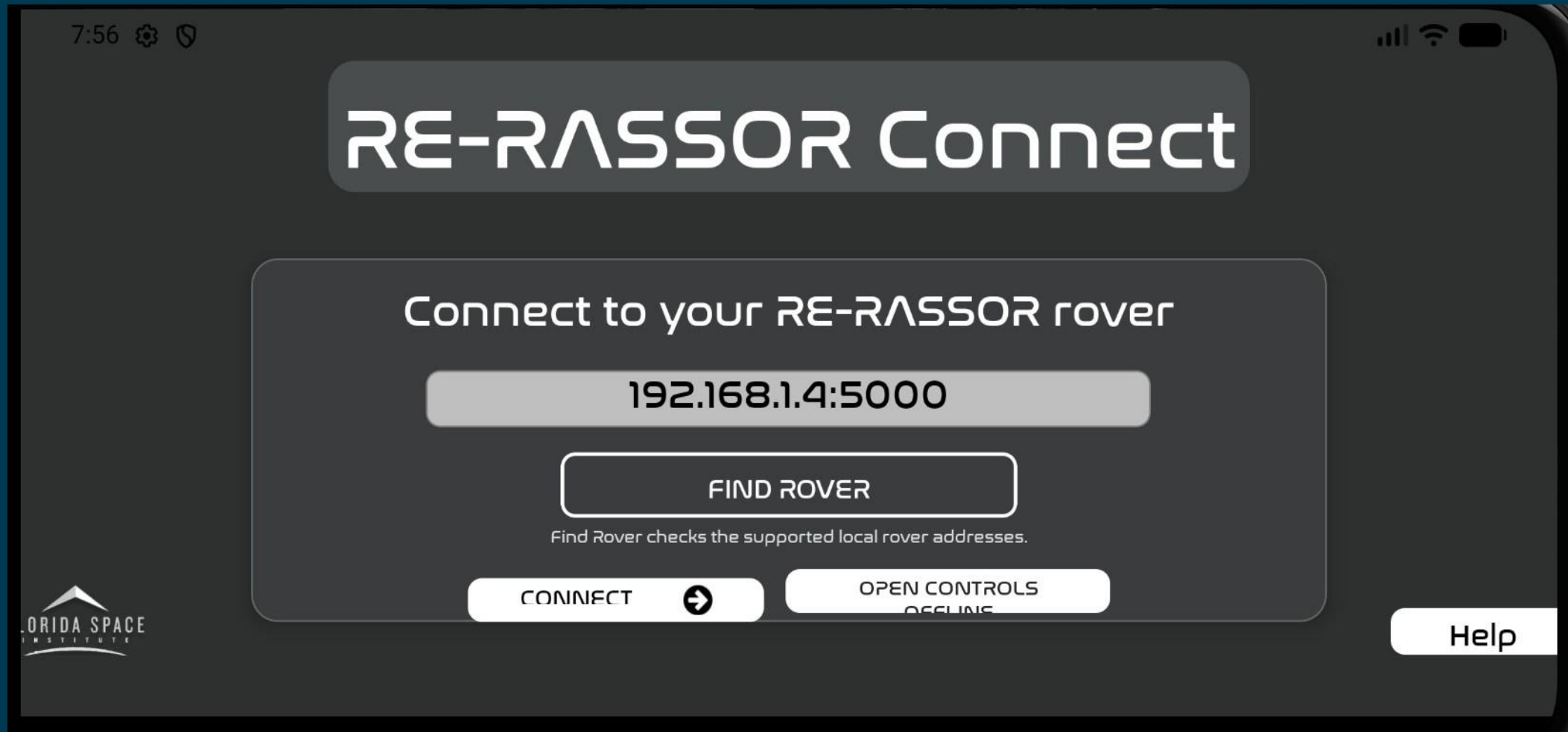
- The central part of the arm is complete and thus far appears to work with the brushless DC motors upon physical manipulation
- The base of the arm is nearly complete with minimal changes needed
- Final initial CAD mock up of the arm is complete and in printing/testing
- The current printed and assembled components have no weight conflicts with the rover resulting in smooth on-rover assembly
- HTTP requests have been drafted and are in testing
- UI is complete with API being developed



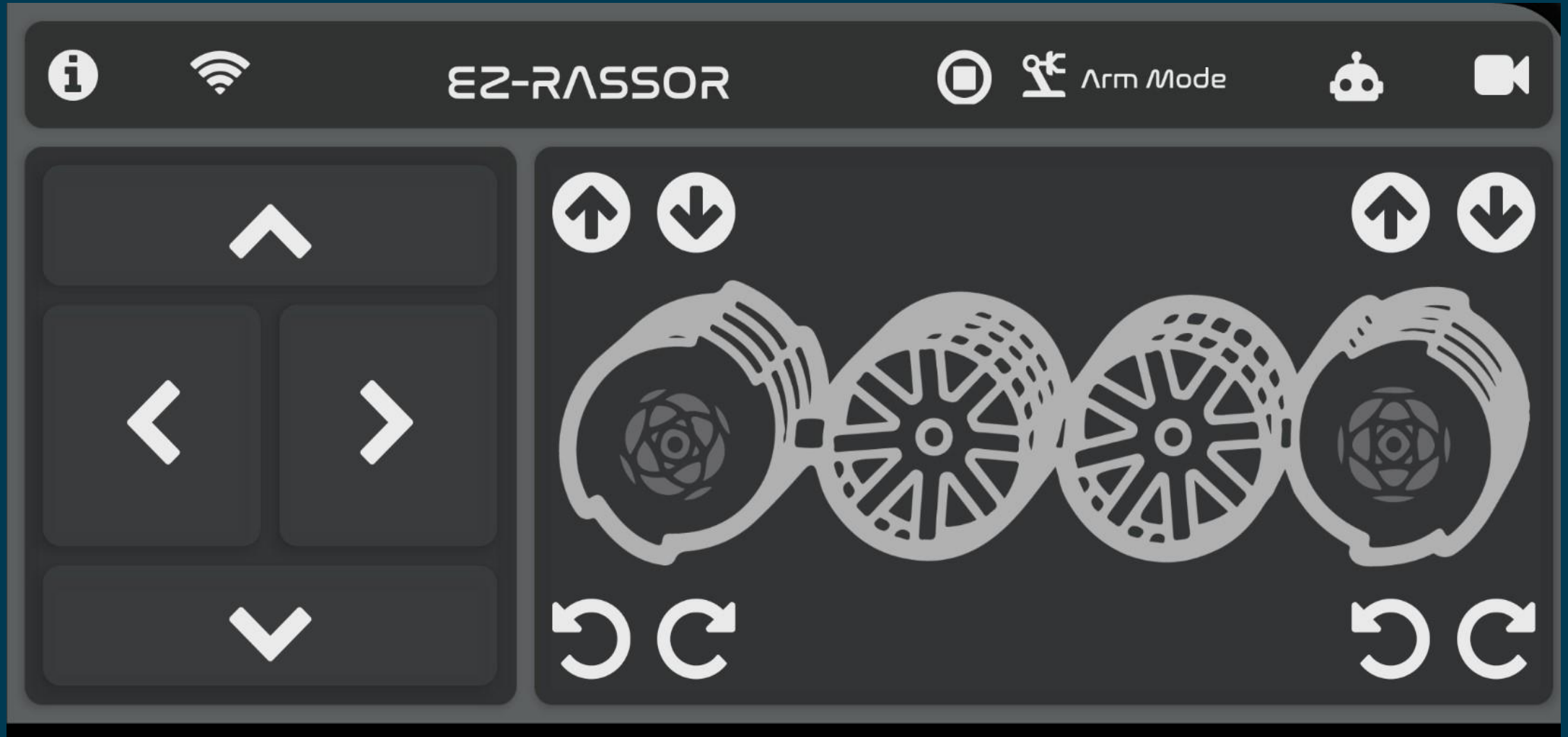
THE CURRENT ROVER



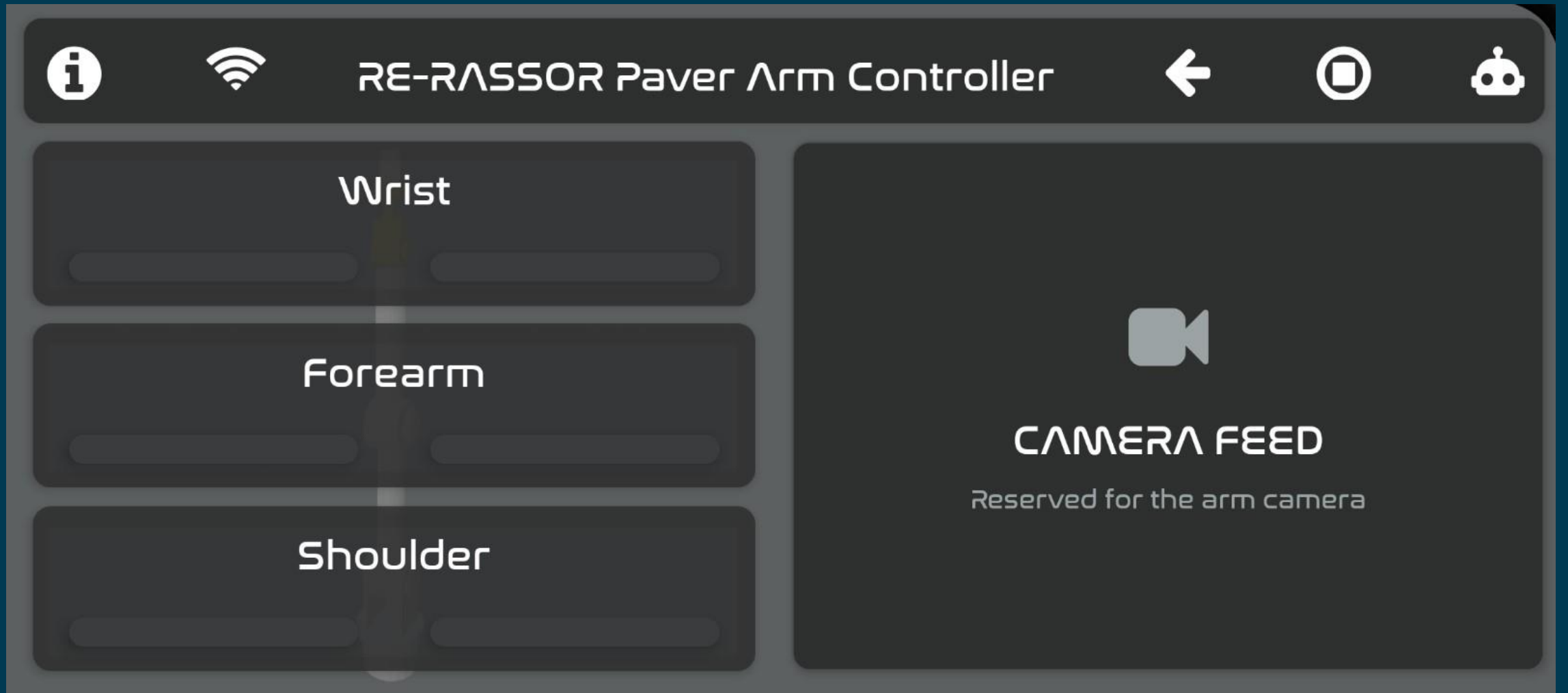
CURRENT UI



CURRENT UI CONT.



CURRENT UI CONT.



CURRENT

HTTP

REQUESTS

- The current system has 2 major forms that commands can take on while being passed around:
 - HTTP messages containing JSON stringified objects
 - Enumerations
- Enumerations are primarily used within the Potato and Arduinos on the rover, while HTTP messages and the corresponding JSON objects are primarily used by the control app to communicate with the rover.
- For example, if the operator chooses to stop the end effector, the corresponding JSON object would be stringified as `"{'end_effector_action':'STOP'}"` and would be posted to the rover in its own HTTP message.
- This would correspond to the `STOP_EFFECTOR` command, enumerated as hexadecimal byte `0x0B`.

OUR CHALLENGES

- The end effector has been complex to design and thus has taken longer to develop than expected
- Overall project complexity was more difficult than expected with a primarily software foundation on hardware project
- Lack of documentation resulted in a steep learning curve for any existing rover features and resulted in initial development delays
- A variety of communication styles and personalities has made organizing and communicating difficult and required a learning curve



Common Causes of Error Include:

- Changes to V2 that resulted in multiple components needing redesign
- Incorrect supports leading to printing issues and difficult to remove components that required redesign
- The printer warping screw holes making exact mechanical connections require multiple adjustments

THE FAILED PROTOTYPES

TOP STRETCH GOALS

Semi-Autonomous Item Identification & Retrieval

- Feasibility Exploration Status: Complete
- Dev / Code Location: PM Branch
- Description:
 - Rover uses the Ultralytics YOLO26 Nano (YOLO26n) model for preliminary item search via the camera and a combination Lucas-Kanade + Canny Algorithm for target tracking after item selection

Interchangeable End Effectors

- Feasibility Exploration Status: To Be Assigned & Complete
- Dev / Code Location: Main / Relevant Member's Branch
- Description:
 - Some lunar rover tasks such as regolith collection are not possible with our base rover arm designs. Interchangeable options would enable future teams to build off our work.

QUESTIONS?

